

Ayush Goel

Senior Software Engineer

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EXPERIENCE

INTUIT

SENIOR SOFTWARE ENGINEER

08/2024 – Present | San Diego, CA

- **Owned the end-to-end design and delivery of mobile driver's license (mDL)** as an account recovery authenticator, partnering with backend and native mobile teams to launch across five states (CA, GA, CO, OH, IL) and expanding supported recovery options for high-assurance users.
- **Designed and implemented Redis caching strategy** for temporal authentication data after identifying it as a reliability and cost risk; reduced DynamoDB write traffic by ~75% in production (98% in pre-production), mitigated customer impact from AWS system errors, and delivered **~\$20K/month in recurring infrastructure savings**.
- **Drove cross-platform design decisions across Identity and Fraud**, aligning risk-based authentication flows while introducing phishing-resistant factors to strengthen account security.
- Identified and eliminated scale-related latency bottlenecks, delivering a **40% reduction in overall service latency** and materially reducing customer-impacting flow incidents.
- **Led cross-team delivery** of Credit Karma sign-in and provisional account initiatives, rapidly ramping into Sign-In Orchestration (SIO) and owning Front Door experience work; unblocked reliability and CI issues that enabled on-time launch and contributed to **\$100M+ in downstream revenue impact**.
- **Set and executed operational reliability strategy** for Credential Service as service owner, migrating production traffic to a dedicated AuthN cluster with zero customer impact, strengthening observability, and helping deliver a full tax season with no major outages across AuthN services.
- **Drove architectural and platform-level improvements across Identity** by identifying systemic reliability and data-consistency risks, implementing preventative fixes (caching for resiliency), and reducing recurring operational and infrastructure costs.
- **Unblocked cross-team delivery by contributing frontend code changes**, leveraging AI-assisted development to prevent dependency bottlenecks and maintain epic timelines.
- **Championed adoption of AI-assisted development workflows**, personally demonstrating productivity gains by compressing multi-person sprint efforts into single sprint delivery; influenced broader team experimentation with AI to increase delivery velocity and quality.
- Served as primary on-call responder for high-severity incidents, leading deep root-cause analysis across service mesh, AWS infrastructure, and cross-region dependencies; prevented repeat incidents through structural fixes rather than point solutions.

SOFTWARE ENGINEER II

07/2022 – 08/2024 | San Diego, CA

- **Service owner for a Tier-1 Credential Service** (Kotlin, Spring Boot), supporting **hundreds of millions of users** via DynamoDB-backed OpenAPI REST and GraphQL APIs; deployed on AWS EKS with ArgoCD-managed canary.
- **Led large-scale Identity 2.0 migration**, ensuring **100% on-time completion** by migrating **620M identities, 2.5B credential records, and 21B user attributes**; personally resolved ~10% of complex edge cases via custom scripts, enabling safe shutdown of legacy 1.0 systems.
- **Reduced infrastructure spend by \$50K/month** by identifying inefficiencies and implementing cost-optimization strategies across Credential Service and eventing pipelines without sacrificing reliability or latency.
- **Improved developer productivity and system reliability** by redesigning CI/CD pipelines (Jenkins), introducing improved autoscaling, and refactoring service architecture to reduce operational burden and failure modes.

SOFTWARE ENGINEERING INTERN

05/2021 – 08/2021 | Remote

- **Designed and implemented CI/CD patterns for Identity 2.0 services**, leveraging ArgoCD, Jenkins, Terraform, Kubernetes, canary deployments, and service mesh to improve deployment safety and confidence.
- **Analyzed deployment metrics across multiple teams** to identify velocity bottlenecks; drove Agile process improvements and delivered proof-of-concepts that increased developer confidence and reduced deployment friction.

SOFTWARE ENGINEERING CO-OP

05/2019 – 12/2019 | San Diego, CA

- Backend engineer on the Identity Platform; **migrated REST services from Spinnaker to Docker/Kubernetes with zero downtime**.
- Built tooling to semi-automate migrations for other teams, reducing duplicated effort and accelerating platform adoption.

EDUCATION

M.S. IN COMPUTER SCIENCE, ROCHESTER INSTITUTE OF TECHNOLOGY
GPA: 4.00

2021 – 2022 | Rochester, NY

B.S. IN COMPUTER SCIENCE, B.S. IN COMPUTATIONAL MATHEMATICS,
ROCHESTER INSTITUTE OF TECHNOLOGY
GPA: 3.79

2017 – 2021 | Rochester, NY

SKILLS

Kotlin • Java • Python • SQL • Groovy • DynamoDB • Redis • Argo CD • git • Spring Boot • Maven • Resilience4j • OpenAPI • GraphQL • Swagger • Wavefront • AWS CDK • Terraform • Gatling • Docker • Kubernetes • Jenkins • Splunk • DynamoDB • Confluence • Tableau • LocalStack • S3 • Databricks